Marine seismology data/metadata standards

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1 Introduction

This document is the result of two years of discussion about how to correctly process and transmit marine seismology data and metadata within the framework of FDSN seismology standards. We present the results here, divided into two sections:

- **Propositions:** for proposed changes to existing FDSN standards and their document, and
- **Recommendations and Reminders:** for recommended processing and preparing methods that will allow instrument facilities to create data and metadata that respects these standards.

A Note on clock corrections

One of the major challenges with distributing marine seismology data is the lack of a timing reference during the deployment period, which can result in time offsets of hundreds of samples by the end of the mission. In the following sections we distinguish between two types of data:

1. **"NOT CLOCK CORRECTED"**: No time correction applied. May be preferred by users of long-period data (>10s) because it can be easier to concatenate daily files.
2. **"CLOCK CORRECTED"**: Indicate the time correction in each record header and apply it. Allows the user to work with time-corrected but otherwise unmodified data.

We recommend providing "CLOCK CORRECTED**"** data.

Some of the recommendations rely on a new (in 2025 or 2026) version of msmod that will allow direct clock drift and leapsecond corrections.

2 Propositions

We propose that the following be added to the following reference standards or their documentation

2.1 Source identifiers

We recommend the following additions to the documentation of FDSN Source Identifiers (https://docs.fdsn.org/projects/source-identifiers/en/v1.0/channel-codes.html#source-and-subsource-codes).

2.1.1 Subsource codes

2.1.1.1 Seismometer channels

If a component's relative angles to another component is known but neither component's absolute angle is known, the angle values should be the relative angles and "plusError" and "minusError" should be set to 180.

For example, for horizontal channels 1 and 2 oriented as recommended below:

- 1: <Azimuth unit="DEGREES" plusError=180 minusError=180> 0 </Azimuth>
- 2: <Azimuth unit="DEGREES" plusError=180 minusError=180> 90 </Azimuth>

We recommend that, if there are two orthogonal horizontal channels that are not known to be aligned to N and E, they be named 1 and 2, with 2 oriented 90° clockwise of 1 when viewed from above.

We recommend that, If the vertical channel is inverted (NOT RECOMMENDED), use subsource code 3.

2.2 StationXML metadata

The following proposals are with respect to StationXML 1.2, documented at https://docs.fdsn.org/projects/stationxml.

2.2.1 New <ClockDrift> element

We propose a new <Station> level <ClockDrift> element, based on <Equipment>. The standard <Equipment> elements describe the instrument's time base. Additional elements are:

<NominalDriftRate> optional

Content_type: double

Nominal drift rate (s/s) of the time base

<Reference> optional

Content type: string

Description of the time reference (i.e. "GNSS" or "cross-correlation with station G:TAM")

<DriftCorrection> optional

If no DriftCorrection is specified, the instrumentation is known to have significant clock drift but this drift has not been measured

Sub-elements:

<Type> required

Allowable values are:

- "PiecewiseLinear": time corrections are linearly interpolated between the <Sync> values
- "CubicSpline": ": time corrections are cubic spline interpolated between the <Sync> values
- "Polynomial {a0} {a1} {a2} [{a3} [{a4} [{a5} …]]]": time corrections are calculated as = new_time = old_time + a0 +a1*deltaT + a2*deltaT**2 + a3*deltaT**3 + …, with deltaT = old_time – the first <Sync><Instrument>.

<Sync> required, many

Instrument-to-reference synchronization. Each sub-element represents the instrument and reference datetimes recorded simultaneously. If there are no Sync elements, the instrumentation is known to have significant clock drift but this drift has not been quantified. Note that there is no extrapolation: for "PiecewiseLinear" and "CubicSpline" types, only times between the first and last <Sync> elements are correctable.

Sub Elements :

<Instrument> required

Content type: datestring (ISO8601 format, followed by "Z" to specify UTC)

All times are assumed to be leapsecond-corrected, if necessary.

`<Reference>` required

Content type: datestring (ISO8601 format, followed by “Z” to specify UTC)

Example:

```
<ClockDrift>
	<Description>MCXO Time Base</Description>
	<Manufacturer>Seascan, Inc</Manufacturer>
	<Model>SISTMB Ver4.1E</Model>
	<NominalDriftRate>1e-8</NominalDriftRate>
	<DriftCorrection>
		<Reference>GNSS</Reference>
		<Type>PiecewiseLinear</Type>
		<Sync>
			<Instrument>2025-01-02T03:00:00Z</Instrument>
			<Reference>2025-01-02T03:00:00Z</Reference>
		</Sync>
		<Sync>
			<Instrument>2025-01-02T03:00:00Z</Instrument>
			<Reference>2025-01-02T03:00:00.034Z</Reference>
		</Sync>
	</DriftCorrection>
</ClockDrift>
```

2.2.2 New <Leveler> element

The instrument's leveling system can affect the quality and absolute values of measurables.

We propose a new <Station> or <Channel> level <Leveler> element, based on <Equipment>. The standard <Equipment> elements describe the leveler hardware. Additional elements are:

`<Threshold>` required

Content type: double

Range: 0.0<= Threshold <=90.0

Deviation from vertical which triggers releveling

Example: <Range>2.0</Range>

Attributes of <Threshold>

unit (type=string, required=no, description=The type of unit being used. This value is fixed to be DEGREES, setting it is redundant.)

`<Accuracy>` required

Content type: double

Range: 0.0<= Accuracy <=90.0

Maximum deviation from vertical accepted after relevel

Example: < Accuracy >0.1</ Accuracy >

Attributes of < Accuracy >

unit (type=string, required=no, description=The type of unit being used. This value is fixed to be DEGREES, setting it is redundant.)

`<MaxRelevelInterval>` optional

Content type: double

Range: 0.0<= MaxRelevelInterval

Longest interval between level checks during the deployment

Example: < MaxRelevelInterval >24</ MaxRelevelInterval >

Attributes of < MaxRelevelInterval >

unit (type=string, required=no, description=The type of unit being used. This value is fixed to be HOURS, setting it is redundant.)

`<NumRelevels>` optional

Content type: integer

Range: 0 <= NumRelevels

Number of relevels during the deployment

Example: < NumRelevels >5</ NumRelevels >

Example:

```
<Leveler>
        <Description>Gimbal with bubble level</Description>
        <Manufacturer>Nanometrics, Inc</Manufacturer>
        <Model>T120 Compact v6.2</Model>
        <Threshold unit=DEGREES>2.0</Threshold>
        <Accuracy unit=DEGREES>0.1</Accuracy>
        <MaxRelevelInterval unit=HOURS>168</MaxRelevelInterval>
</Leveler>
```

2.3 miniSEED data

The following are recommended additions to miniSEED documentation (https://www.fdsn.org/pdf/SEEDManual_V2.4.pdf and https://fdsn-docs.readthedocs.io/projects/miniseed3/en/latest/index.html). “FSDH” refers to the miniSEED 2/3 Fixed Section Data Header. “EH” refers to a miniSEED Extra Header with the specified path

2.3.1 Clock drift correction

(Note that the Time Correction is ADDED to the original instrument time.)

The miniSEED3 rules are direct translations of the miniSEED 2 rules following the recommended mapping from miniSEED 2.4.

"NOT CLOCK CORRECTED" data:

- miniSEED2::
    i. Set the FSDH data quality indicator to "D".
    ii. Verify that FSDH activity flags (field 12) bit 1 (“time correction applied”) equals 0
- miniSEED3:
    i. Set the FSDH Data publication version to 2
    ii. Verify that EH.FDSN.Time.Correction equals “0” or does not exist

"CLOCK CORRECTED" data: miniSEED2:

iii. Set the FSDH data quality indicator to "Q".
iv. Set FSDH field 16 (“Time Correction”), to the correction (in units of 0.0001 seconds).
v. Set FSDH activity flags (field 12) bit 1 (“time correction applied”) to 1.

- miniSEED3:
    i. Set the FSDH Data publication version to 3
    ii. Set EH.FDSN.Time.Correction to the correction (text, units of float seconds)

Note: the msmod command “--cc” ,recommended for correcting clock drifts, correctly sets the record header “Time Correction” (and, for miniSEED2, “time correction applied”) fields.

2.3.2 Unmeasured clock drift

If clock drift is expected, but not measured/estimated:

- miniSEED2:
    - Set the FSDH data quality indicator to "D"
    - Set the FSDH data quality flags (field 14), bit 7 ("time tag is questionable") to 1.
- miniSEED3:
    - Set the FDSH Data publication version to 2
    - Set the FSDH Flags (field 3) bit 1 ("time tag is questionable") to 1

2.3.3 Leap second correction

Leap seconds should be corrected in **CLOCK CORRECTED** data.

Note: The msmod “--ls” command allows you to add leapsecond(s) to data that were synchronized to leapseconds at deployment time but not afterwards.

3 Recommendations and Reminders

The following recommendations and reminders are intended to help collectors of marine seismology data and manufacturers of dataloggers to prepare their data and metadata so that they can be easily distributed and used by other researchers. Following these recommendations should allow the greatest compatibility with seismology data from land stations and from marine stations collected by other groups.

3.1 Source identifiers

The following are source identifier rules for pressure sensors:

- Set Dip=90. if the recorded value DECREASES for a pressure increase.
- Set Dip=-90. if the value INCREASES for a pressure increase.
- Set the subsource code to "H" if the sensor is a hydrophone
- Set the subsource code to "G" if the sensor is a differential pressure gauge
- Set the subsource code to "O" if the sensor is an absolute pressure gauge / bottom pressure recorder

3.1.1 Station names for repeated deployments

The IASPEI Station Coding Standard recommends that station and/or location codes be changed if the associated sensors are moved far enough to result in a significant ***teleseismic*** travel-time residual discrepancy. We recommend the same, except that the basis for what is a significant travel-time residual discrepancy should depend on your experiment.

If you change the station name between deployments, we recommend incrementing the last *N* characters of the station name, e.g. ``STAA``, ``STAB``, ``STAC``, or ``STA01``, ``STA02``, ``STA03``, etc. The value of *N* depends on the maximum number of deployments you will possibly make and the characters you use in the incrementor (numeric, alphabetic, or alphanumeric).

3.2 StationXML metadata

3.2.1 Subnetwork files

We recommend using obsinfo subnetwork files to store essential information about OBS deployments. These files can help in creating StationXML files containing this information.

3.2.2 Adding non-standard information

If you have information that you consider important to add to metadata, you should propose a new element for stationXML. Exceptions are information that changes frequently throughout the deployment (see if it can fit into the miniSEED file, the proposed GeoJSON format or the under-

discussion Annotation format). If your information is relevant to StationXML and your proposal has not been acted on, you can embed the information

- ~~as a JSON formatted string in a `<Comment><Value>` element or a `<Equipment><Description>` element. The JSON formatted string should follow your proposed future element's format.~~
- inside a `<Comment><Value>` element as `<![CDATA[proposed_element]]>`

Below is an example of embedding the proposed `<ClockDrift>` element:

```
<Comment>
    <Value><![CDATA[<ClockDrift>
            <Description>MCXO Time Base</Description>
            <Manufacturer>Seascan, Inc</Manufacturer>
            <Model>SISTMB Ver4.1E</Model>
            <NominalDriftRate>1e8</NominalDriftRate>
            <DriftCorrection>
                <Reference>GNSS</Reference>
                <Type>PiecewiseLinear</Type>
                <Sync>
                    <Instrument>2025-01-02T03:00:00Z</Instrument>
                    <Reference>2025-01-02T03:00:00Z</Reference>
                </Sync>
                <Sync>
                    <Instrument>2025-01-02T03:00:00Z</Instrument>
                    <Reference>2025-01-02T03:00:00.034Z</Reference>
                </Sync>
            </DriftCorrection>
        </ClockDrift>]]>
    </Value>
</Comment>
```

3.2.3 Input units

For pressure sensor channel responses, `InputUnits` should be `Pa`, without a `Description`

3.2.4 Deployments in lakes

We remind that the `<WaterLevel>` elements should be set to the elevation of the lake surface

3.2.5 Uncertainty in station/channel positions, dips and azimuths

We recommend using the `plusError`, `minusError` and `measurementMethod` attributes to specify uncertainties in `Latitude`, `Longitude` and `Elevation`, plus estimated `Dip` and `Azimuth`, and how these values were measured/estimated.

3.3 miniSEED data

3.3.1 Data completeness

We recommend keeping all of the recorded data, including "noisy" or "bad" sections.

3.3.2 Correcting drift in data that contains a leap second

msmod (version 2026+) can add leap-seconds to data, see its help.

"**NOT CLOCK CORRECTED**" data should not be leap-second corrected

"**CLOCK CORRECTED**" data MUST be leap-second corrected

We suggest that the drift correction be performed before the leap-second correction, to avoid the possibility of the leap-second being shifted outside of the record with the leap-second flag.

Exception: If you use the data to calculate the drift rate, first generate LEAP-CORRECTED data from the NOT CLOCK CORRECTED DATA. Use this data to calculate the clock drift, then throw the LEAP-CORRECTED data away and process as suggested above.

3.3.3 Sample rate in miniSEED3

Specify the actual sample rate (for example 249.9999999923 sps) in each FSDH (this is not possible in miniSEED2, which can only use 16- or 32-bit words for the sample rate)

3.4 Other recommendations

3.4.1 Geophone wiring and polarity

The default polarity on short-period geophones manufactured for the exploration industry (e.g. the Geospace GS-11D) is such that a positive output voltage represents downward ground motion, in contradiction to the standard followed in passive seismology where a positive output voltage represents upward ground motion. We recommend that all ground-motion sensors follow the passive seismology standard. For geophones, this can be readily accomplished by connecting the positive (negative) terminal of the geophone to the negative (positive) inputs to the data-logger. This can be done on the geophone itself or by switching the leads from the geophone at the data-logger inputs. Alternatively, some data-loggers allow switching polarity in software.

3.4.2 Saving processing information

We recommend that processing done on data files (from data download to delivery to the data center) should be recorded in text-based, structured files. The JSON process-steps format is an example.